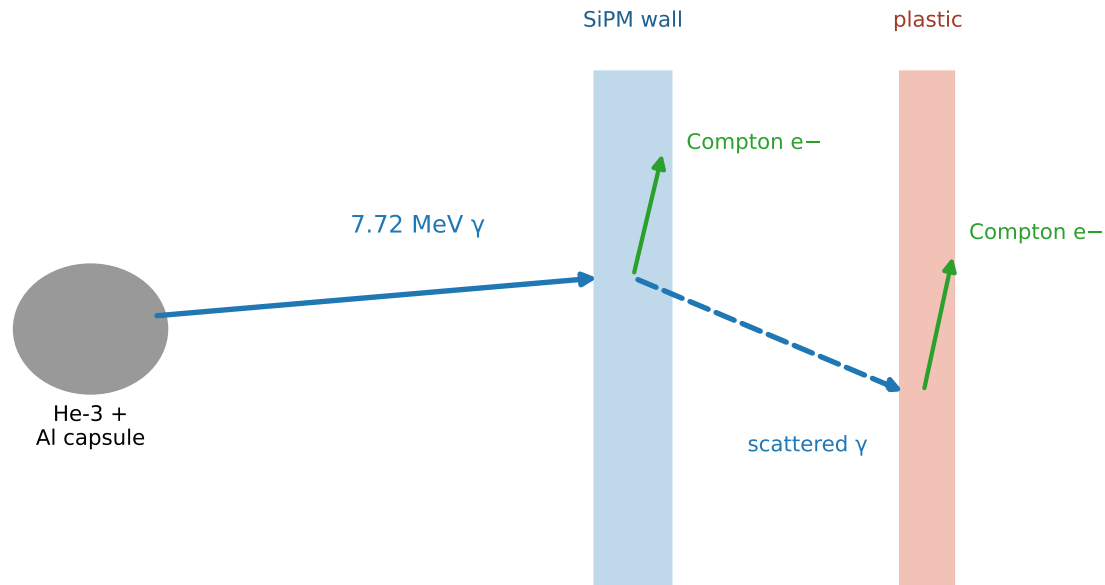
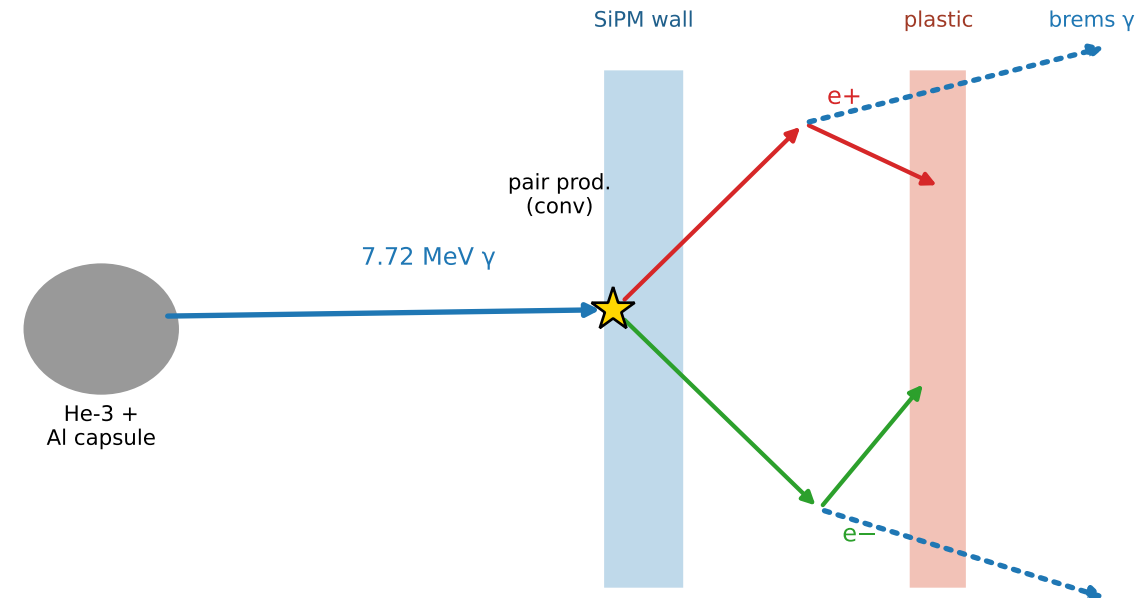


How the $^{27}\text{Al}(n,\gamma)$ 7.72 MeV capture γ makes a SiPM $\wedge$ plastic leg (both deposit >0.5 MIP in one arm)

(1) Double Compton — the typical leg
 γ Compton-scatters in SiPM, the scattered γ Compton-scatters in plastic



(2) Pair production + EM shower — the high-energy leg
7.72 MeV γ converts (mostly in the Al capsule); $e^\pm$ spray both detectors



Detector interactions are Compton-dominated (thin low-Z); pair production happens mainly in the denser Al capsule and dominates the highest-energy legs. The soft 2.22 MeV H-capture γ rarely clears both thresholds.